

RepSense AI - Output Data Sample

Hackathon submission output files

Main output: output/AI_rep_management_output_sample.csv | **Rows:** 8 | **Columns:** 45 | **Granularity:** territory and analysis date.

Purpose: The output contains the territory risk analysis produced by the pipeline, including ML anomaly scores, business-rule scores, severity, explanations, and recommendations.

Main Risk Output - Key Fields

date	territory id	sales	transactions	active reps	hcs	sales change pct	avg activity	avg hcp coverage	current reps	required reps	vacancies	promotion count	isolation forest score	lof score	ensemble anomaly score	sales risk score	crm risk score	hcp coverage risk score	vacancy risk score	ml risk score	business risk score	final risk score	severity	risk reasons	ai recommendation
2017-01-01	TER_R_01	1550393976.312982	32018	13	94	0.0	82.99144092878134	1.1164601044760831	4	2	0	6	100.0	0.0	60.0	0.0	0.0	25.0	0.0	60.0	5.0	23.5	Minor	HCP coverage 1% (below 85%); Isolation Forest de	Continue routine monitoring. Minor risks detected
2017-01-01	TER_R_02	1504139008.51158	32252	13	95	0.0	82.66593975599497	1.177916623988528	0	2	2	4	42.58416147768629	36.92997904495101	40.32248850459217	0.0	0.0	25.0	60.0	40.32248850459217	17.0	22.61287097660726	Minor	2 rep vacancy/vacancies; HCP coverage 1% (below	Continue routine monitoring. Minor risks detected
2017-01-01	TER_R_03	1602690127.112361	32702	13	95	0.0	82.11602046890019	1.0037898187032674	0	2	2	8	65.37963362260709	57.932196366228204	62.40065872005553	0.0	0.0	25.0	60.0	62.40065872005553	17.0	30.340230552019435	Minor	2 rep vacancy/vacancies; HCP coverage 1% (below	Continue routine monitoring. Minor risks detected
2017-01-01	TER_R_04	1486118343.67167	32087	13	94	0.0	80.41761556806587	0.9218477926866744	1	2	1	7	51.453714744390254	82.0032673867533	63.67353580133547	0.0	0.0	25.0	25.0	63.67353580133547	10.0	27.28573753046741	Minor	1 rep vacancy/vacancies; HCP coverage 1% (below	Continue routine monitoring. Minor risks detected
2017-01-01	TER_R_05	1345784287.46154	31634	13	94	0.0	82.8304674784864	0.9525760524428968	1	2	1	4	83.25305563190571	57.932196366228204	73.1247119256347	0.0	0.0	25.0	25.0	73.1247119256347	10.0	30.593649173972143	Minor	1 rep vacancy/vacancies; HCP coverage 1% (below	Continue routine monitoring. Minor risks detected
2017-01-01	TER_R_06	1407142022.521303	30702	13	93	0.0	81.21500381584511	1.0652463382157125	2	2	0	7	70.32167071193322	89.08999722078168	77.82900131547261	0.0	0.0	25.0	0.0	77.82900131547261	5.0	29.74015046041541	Minor	HCP coverage 1% (below 85%)	Continue routine monitoring. Minor risks detected
2017-01-01	TER_R_07	1390167904.038781	31531	13	93	0.0	83.22385635362807	0.9320905459387484	1	2	1	9	94.8587317637276	2.2835001674381226e-13	56.91523905823664	0.0	0.0	25.0	25.0	56.91523905823664	10.0	24.920333670382824	Minor	1 rep vacancy/vacancies; HCP coverage 1% (below	Continue routine monitoring. Minor risks detected

date	territory id	sales	transactions	active reps	hcps	sales change pct	avg activity	avg hcp coverage	current reps	required reps	vacancies	promotion count	isolation forest score	lof score	ensemble anomaly score	sales risk score	crm risk score	hcp coverage risk score	vacancy risk score	ml risk score	business risk score	final risk score	severity	risk reasons	ai recommendation
2017-01-01	TER_R_08	1512542072.5652	31152	13	93	0.0	81.39733932793814	1.0140325719553416	1	2	1	5	0.0	100.0	40.0	0.0	0.0	25.0	25.0	40.0	10.0	19.0	Minor	1 rep vacancy/vacancies; HCP coverage 1% (below	Continue routine monitoring. Minor risks detected

Output File Inventory

File	Path	Rows	Columns
Main risk output	output/AI_rep_management_output_sample.csv	8	45
HR and vacancy data	output/rep_master_hr_sample.csv	13	8
CRM activity data	output/crm_activity_detail_sample.csv	104	12
Promotional activity	output/promotional_activity_sample.csv	50	12
Executive summary	output/executive_dashboard_sample.csv	1	6
Critical territory filter	output/critical_territories_sample.csv	0	45

HR and vacancy data

File: output/rep_master_hr_sample.csv | Rows: 13 | Columns: 8

rep id	name of sales rep	manager	sales team	primary territory	employment status	vacancy flag	vacancy days
REP_001	Abigail Thompson	Tracy Banks	Bravo	TERR_01	Active	0	0
REP_002	Alan Ray	James Goodwill	Alfa	TERR_04	Exit	0	0
REP_003	Anne Wu	Britanny Bold	Delta	TERR_07	Active	0	0
REP_004	Daniel Gates	Alisha Cordwell	Charlie	TERR_01	Active	0	0
REP_005	Erica Jones	James Goodwill	Alfa	TERR_01	Active	0	0
REP_006	Jessica Smith	Britanny Bold	Delta	TERR_01	Active	0	0
REP_007	Jimmy Grey	Alisha Cordwell	Charlie	TERR_06	Active	0	0
REP_008	Mary Gerrard	Britanny Bold	Delta	TERR_07	Vacant	1	99
REP_009	Morris Garcia	Tracy Banks	Bravo	TERR_08	Active	0	0
REP_010	Sheila Stones	Britanny Bold	Delta	TERR_04	Active	0	0
REP_011	Stella Given	Alisha Cordwell	Charlie	TERR_05	Active	0	0
REP_012	Steve Pepple	Tracy Banks	Bravo	TERR_06	Active	0	0

CRM activity data

File: output/crm_activity_detail_sample.csv | Rows: 104 | Columns: 12

date	rep id	name of sales rep	territory id	planned calls	actual calls	planned visits	actual visits	call completion pct	visit completion pct	hcp coverage count	hcp coverage pct
2017-01-01	REP_008	Mary Gerrard	TERR_07	87	64	58	42	73.5632183908046	72.41379310344827	11	1.4647137150466043
2017-01-01	REP_006	Jessica Smith	TERR_01	84	62	36	26	73.80952380952381	72.22222222222221	6	0.7989347536617843
2017-01-01	REP_012	Steve Pepple	TERR_06	63	48	61	47	76.19047619047619	77.04918032786885	5	0.6657789613848202
2017-01-01	REP_008	Mary Gerrard	TERR_05	100	89	44	39	89.0	88.63636363636364	3	0.3994673768308922
2017-01-01	REP_003	Anne Wu	TERR_07	120	85	51	36	70.83333333333334	70.58823529411765	5	0.6657789613848202
2017-01-01	REP_013	Thompson Crawford	TERR_05	83	60	37	26	72.28915662650603	70.27027027027027	9	1.1984021304926764
2017-01-01	REP_010	Sheila Stones	TERR_04	109	89	65	53	81.65137614678899	81.53846153846153	9	1.1984021304926764
2017-01-01	REP_008	Mary Gerrard	TERR_03	82	61	79	58	74.39024390243902	73.41772151898735	10	1.3315579227696404
2017-01-01	REP_011	Stella Given	TERR_05	152	141	91	84	92.76315789473684	92.3076923076923	7	0.9320905459387484
2017-01-01	REP_009	Morris Garcia	TERR_08	99	78	63	49	78.78787878787878	77.77777777777779	6	0.7989347536617843
2017-01-01	REP_005	Erica Jones	TERR_01	81	72	52	46	88.88888888888889	88.46153846153845	6	0.7989347536617843

date	rep id	name of sales rep	territory id	planned calls	actual calls	planned visits	actual visits	call completion pct	visit completion pct	hcp coverage count	hcp coverage pct
2017-01-01	REP_001	Abigail Thompson	TERR_01	81	71	44	38	87.65432098765432	86.36363636363636	9	1.1984021304926764

Promotional activity

File: output/promotional_activity_sample.csv | Rows: 50 | Columns: 12

date	rep id	name of sales rep	territory id	hcp id	product name	product class	promotion type	sales before	sales after	sales uplift %	promotion impact
2017-01-01	REP_010	Sheila Stones	TERR_06	HCP_00660	Relenvice	Antimalarial	HCP Meeting	18592.664926604462	25574.32015794336	37.55058921837912	High Impact
2017-01-01	REP_008	Mary Gerrard	TERR_02	HCP_00093	Alemtuvatol Megalinum	Analgesics	Campaign	9232.697336099736	10569.91666939058	14.483517488029548	Medium Impact
2017-01-01	REP_011	Stella Given	TERR_07	HCP_00393	Aggrakine	Antiseptics	Product Presentation	22992.650633185905	29164.69523259689	26.84355404636432	High Impact
2017-01-01	REP_001	Abigail Thompson	TERR_05	HCP_00417	Interzide	Antimalarial	Educational Session	7761.870629119898	8590.099926686671	10.67048572620546	Medium Impact
2017-01-01	REP_008	Mary Gerrard	TERR_04	HCP_00647	Aquamycin Lacoran	Analgesics	Campaign	19255.865018576587	23770.564969872947	23.44584336741517	High Impact
2017-01-01	REP_001	Abigail Thompson	TERR_08	HCP_00103	Angioparin Brimosumab	Mood Stabilizers	Product Presentation	17771.301495105618	24037.69507454214	35.26130925842628	High Impact
2017-01-01	REP_008	Mary Gerrard	TERR_06	HCP_00195	Adreacetam Barazoxane	Antimalarial	Educational Session	22853.507343360936	30629.78668437321	34.02663417994493	High Impact
2017-01-01	REP_009	Morris Garcia	TERR_03	HCP_00733	Benzoprox	Antiseptics	Campaign	7135.738160063239	9299.23451496932	30.31916679643002	High Impact
2017-01-01	REP_010	Sheila Stones	TERR_01	HCP_00105	Symdocet	Antipiretics	Educational Session	1124.3405901093884	1304.4845262491965	16.02218560145387	Medium Impact
2017-01-01	REP_009	Morris Garcia	TERR_03	HCP_00485	Decidenu Galantrigine	Antipiretics	Product Demonstration	1831.5358259779996	2546.2172956370355	39.02088397738069	High Impact
2017-01-01	REP_010	Sheila Stones	TERR_03	HCP_00539	Aspinavir Silovance	Antibiotics	Campaign	14591.559716130498	17448.24495349248	19.577655116635764	Medium Impact
2017-01-01	REP_001	Abigail Thompson	TERR_05	HCP_00666	Choriotrisin	Antibiotics	HCP Meeting	11829.227771663487	15613.314760270134	31.98929855481603	High Impact

Executive summary

File: output/executive_dashboard_sample.csv | Rows: 1 | Columns: 6

date	total sales	territory count	avg risk score	total vacancies	critical count
2017-01-01	11798977742.195415	8	25.99912154548306	8	0

Critical territory filter

File: output/critical_territories_sample.csv | Rows: 0 | Columns: 45

date	territory id	sales	transactions	quantity	customers	products	active reps	hcps	previous month sales	sales change pct	rolling 3m sales	sales vs 3m avg pct	planned calls	actual calls	planned visits
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Risk Interpretation

The final risk score is normalized from 0 to 100. Minor covers 0-39.99, Major covers 40-69.99, and Critical covers 70-100. The score combines ML anomaly risk with sales, CRM, HCP coverage, vacancy, and long-vacancy components. A formal accuracy percentage is not available because the current anomaly models are unsupervised and no historical confirmed-risk labels are present.

Submission folder: output/